

1 Bias and Power of Slope Versus Final-Visit Estimands Under
2 Nonlinear Treatment Effects in Alzheimer’s Disease Trials

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33 Abstract

34 We use Monte Carlo simulation to quantify how nonlinearity in the treatment effect tra-
35 jectory affects the bias, coverage, and power of a random-slopes mixed model relative to a
36 categorical-time MMRM, in the setting of an 18-month Alzheimer’s disease trial with CDR-
37 SB as the endpoint and a fixed total treatment effect of -0.45 points at month 18 (matching
38 CLARITY-AD). Across seven values of a shape parameter κ spanning early-onset to delayed-
39 onset treatment effect trajectories, the random-slopes model is approximately unbiased only
40 near $\kappa = 1$ (exact linearity) and its bias grows to 0.225 points (versus a true effect of -0.45)
41 at the most convex trajectory examined ($\kappa = 0.3$); its 95% CI coverage falls as low as 0.827
42 in the same condition. The categorical-time MMRM remains approximately unbiased and
43 maintains coverage at or above 0.941 across all conditions examined. Contrary to the conven-
44 tional expectation that a correctly specified parsimonious mean structure is more efficient, the
45 unconstrained random-slopes model does not achieve higher power than the categorical-time
46 MMRM even under exact linearity in this design; we show this is attributable in part to an
47 estimator choice (an unidentified treatment main effect retained in a model fit to post-baseline
48 data only) rather than to the linearity assumption per se. A full-scale ($n_{\text{rep}} = 1000$ per κ)
49 evaluation of a constrained alternative that removes this main effect closes the power gap with
50 the categorical-time MMRM under exact linearity (constrained power 0.678 versus 0.619 at
51 $\kappa = 1$), while a separate null-scenario simulation confirms all three estimators control type I
52 error near the nominal 0.05 level.

53 **Keywords:** mixed model for repeated measures; random slopes; Alzheimer’s disease; CDR-
54 SB; estimand; simulation study; nonlinearity.

55 Data and code availability

56 The simulation driver (`analysis/scripts/sim_mmr_linear.R`), the report source
57 (`analysis/report/report.Rmd`), and this compendium’s `renv.lock` are version-controlled
58 in the `mmrmrobust` repository. `analysis/scripts/run_sim_08.R` reproduces the cached
59 simulation output loaded by this report (`analysis/data/derived_data/sim_08.rds`); see
60 the Reproducibility statement at the end of this document for the exact command and RNG
61 settings used.

62 1 Introduction

63 1.1 The CDR-SB as a primary endpoint in AD trials

64 The Clinical Dementia Rating Sum of Boxes (CDR-SB) has become the primary efficacy
65 endpoint for disease-modifying trials in early Alzheimer’s disease (AD). The instrument in-
66 tegrates cognition and function into a single composite score ranging from 0 to 18, and it

67 demonstrates superior responsiveness to change in early-stage populations compared with tra-
68 ditional cognitive scales such as the ADAS-Cog^{1,2}. A parallel line of work has instead built
69 composite cognitive endpoints for the preclinical stage, such as the Preclinical Alzheimer Cog-
70 nitive Composite³; CDR-SB remains the dominant choice once clinical impairment is present,
71 which is the setting considered here. Annual CDR-SB progression rates in placebo arms of
72 recent trials are approximately 1.3 to 1.7 points per year for participants with baseline CDR
73 global scores of 0.5 to 1.0^{1,4}.

74 1.2 Treatment effects from recent anti-amyloid trials

75 The approval of lecanemab and the positive results for donanemab have provided the first evi-
76 dence that amyloid-targeting therapies can slow clinical decline on CDR-SB. In the CLARITY-
77 AD trial, lecanemab reduced 18-month CDR-SB worsening by 0.45 points (27% slowing) rel-
78 ative to placebo, with the treatment arm declining 1.21 points versus 1.66 in placebo⁵. The
79 TRAILBLAZER-ALZ 2 trial of donanemab demonstrated a similar magnitude of slowing on
80 CDR-SB in participants with low/medium tau burden⁶. By contrast, the earlier Phase 3 stud-
81 ies of aducanumab did not show a consistent CDR-SB benefit across trials⁷, underscoring that
82 both the magnitude and the shape of the treatment-effect trajectory are empirical questions
83 rather than assured features of amyloid-targeting therapy. In the two positive trials, inspec-
84 tion of the CDR-SB trajectories over time reveals a non-constant treatment difference: the
85 separation between arms widens gradually, with the rate of separation appearing to change
86 across the 18-month follow-up period.

87 1.3 The linearity question

88 Both regulatory agencies and the statistical community have debated the choice between two
89 estimands for these trials:

- 90 • **Final-visit estimand:** The between-group difference in mean change from baseline
91 at the last scheduled visit (e.g., month 18). This corresponds to the categorical-time
92 MMRM, which places no parametric constraint on how the mean response evolves over
93 time⁸.
- 94 • **Slope estimand:** The between-group difference in the rate of change per unit time.
95 This corresponds to a random-slopes mixed model that assumes the treatment effect
96 accumulates linearly over the follow-up period.

97 The U.S. Food and Drug Administration’s 2018 draft guidance on early AD drug development
98 discusses the use of slope analyses as a potentially more sensitive approach for detecting
99 disease-modifying effects, particularly when the mechanism is expected to produce a constant
100 rate of slowing⁹. The European Medicines Agency’s guideline similarly acknowledges slope-
101 based endpoints for disease-modification claims, while also noting that the assumption of
102 constant treatment effect rates must be verified¹⁰. The ICH E9(R1) addendum on estimands
103 provides the formal framework for distinguishing these two targets of estimation, emphasizing
104 that the choice of estimand should precede the choice of statistical model¹¹.

1.4 Why linearity may not hold

Several biological and design factors suggest that the treatment effect in AD trials is unlikely to be perfectly linear in time:

1. **Delayed pharmacodynamic onset.** Anti-amyloid antibodies require weeks to months to achieve meaningful amyloid clearance. The clinical benefit may lag behind the biological effect, producing a delayed-onset or concave treatment effect trajectory (the effect accelerates over time).
2. **Plateau or diminishing returns.** Conversely, if the treatment primarily benefits patients at earlier disease stages, the marginal benefit per additional month of treatment could decrease as disease progresses, producing a convex (early-onset) trajectory.
3. **Nonlinear placebo progression.** Even if the treatment modifies the underlying rate by a constant fraction, nonlinearity in the natural disease trajectory (documented by¹² using nonlinear mixed-effect models) would produce a nonlinear absolute treatment difference over time.

Despite the clinical relevance of this question, limited simulation evidence exists quantifying how much nonlinearity the random-slopes model can tolerate.¹³ compared the slope model with the categorical-time MMRM for ADAS-Cog data, finding that the slope model is more powerful when linearity approximately holds, but biased otherwise. That study, however, did not systematically vary the degree of nonlinearity, nor did it focus on CDR-SB.

1.5 Present study

We use Monte Carlo simulation to evaluate the robustness of a random-slopes mixed model (a linear mixed-effects model, not itself an MMRM in the clinical-trials sense; see Methods) to departures from linearity in the treatment effect trajectory. We simulate 18-month AD trials with CDR-SB as the primary endpoint, visits at 3-month intervals, and a total treatment effect of -0.45 points at month 18 (matching CLARITY-AD). We parameterize the shape of the treatment effect trajectory using a power function $g(t) = (t/t_{\max})^\kappa$, where $\kappa = 1$ yields a linear effect, $\kappa < 1$ yields an early-onset (convex) pattern, and $\kappa > 1$ yields a delayed-onset (concave) pattern. We compare three analysis models:

1. An **unconstrained random-slopes model** that fits treatment-by-time as a linear interaction (retaining a treatment main effect) and extracts the treatment contrast at 18 months from the interaction term alone.
2. A **constrained random-slopes model** that omits the treatment main effect, imposing equality of the two arms at baseline.
3. A **categorical-time MMRM** that fits treatment-by-visit as an unstructured interaction and extracts the treatment contrast at the final visit.

We evaluate bias, empirical standard error, model-based standard error, power (proportion of $p < 0.05$), and 95% confidence interval coverage across a range of κ values, and additionally report empirical type I error for all three models in a separate null-scenario ($\delta = 0$) simulation.

2 Methods

2.1 ADEMP structure

Following Morris, White, and Crowther¹⁴, the simulation is reported under the ADEMP framework.

- **Aims.** Quantify the robustness of a random-slopes model fit (which imposes linear-in-time treatment effects) to nonlinear treatment trajectories governed by the shape parameter κ .
- **Data-generating mechanism.** A longitudinal model where the treatment effect accrues as $g(t, \kappa) = (t/t_{\max})^\kappa$, with κ varied across $\{0.3, 0.5, 0.7, 1.0, 1.5, 2.0, 3.0\}$.
- **Estimand.** The total treatment effect at month 18, $\delta = -0.45$, which is invariant to κ by construction.
- **Methods.** (i) Unconstrained random-slopes model via `nlme::lme`, which assumes a linear-in-time treatment effect and retains a treatment main effect; (ii) constrained random-slopes model via `nlme::lme`, omitting the treatment main effect; (iii) categorical-time MMRM via the `mmrm` package with unstructured covariance.
- **Performance measures.** Convergence rate, bias, empirical SE, mean model-based SE, power at $\alpha = 0.05$, and coverage of 95% CIs, each reported with its Monte Carlo SE per Morris Table 6; a separate null-scenario ($\delta = 0$) simulation additionally reports empirical type I error for all three models.

2.2 Data-generating model

We generate data for $N = 200$ participants per arm (total $N = 400$) observed at months $t \in \{0, 3, 6, 9, 12, 15, 18\}$. The outcome for participant i at visit j is:

$$Y_{ij} = \beta_0 + b_{0i} + (\beta_1 + b_{1i})t_j^* + \delta \cdot g(t_j, \kappa) \cdot \text{trt}_i + \varepsilon_{ij}$$

where $t_j^* = t_j/12$ is time in years, trt_i is a binary treatment indicator, and:

$$g(t_j, \kappa) = \left(\frac{t_j}{t_{\max}} \right)^\kappa$$

The random effects (b_{0i}, b_{1i}) follow a bivariate normal distribution with standard deviations $\sigma_{b_0} = 1.0$ and $\sigma_{b_1} = 0.6$ (in CDR-SB points per year) and correlation $\rho = 0.3$. The within-subject errors ε_{ij} have an AR(1) structure with marginal variance $\sigma^2 = 1.44$ and autocorrelation $\phi = 0.4$.

2.3 Parameter values

The fixed-effect parameters are calibrated to approximate the placebo arm of CLARITY-AD:

Parameter	Value	Interpretation
β_0	3.2	Mean baseline CDR-SB
β_1	1.5 points/year	Placebo annual decline
δ	-0.45	Treatment effect at $t_{\max}$
κ	Varies	Shape of treatment effect

The κ parameter takes values in $\{0.3, 0.5, 0.7, 1.0, 1.5, 2.0, 3.0\}$, spanning strongly convex (early-onset) to strongly concave (delayed-onset) treatment effect trajectories.

2.4 Analysis models

Both models are fit to post-baseline observations only (months 3 through 18); neither model adjusts for baseline CDR-SB, which departs from the near-universal practice of baseline-adjusted change-from-baseline MMRM⁸. We disclose this as a design choice made to keep the two models on comparable footing (a categorical model with visit 3 as the reference level already absorbs baseline-like information from the first post-randomization visit); it is not evaluated as a sensitivity condition here and is noted as a limitation below.

Random-slopes model (unconstrained). We fit a linear mixed-effects model (via `nlme::lme`) with fixed effects for treatment, time (continuous, in years), and their interaction, plus random intercepts and random slopes for time. Because the fixed-effect specification includes a treatment main effect while the model is fit to post-baseline data only (excluding $t = 0$, where the true treatment effect is 0 by randomization), the main effect is not anchored to an observed baseline difference. The treatment effect at 18 months is extracted as $\hat{\beta}_{\text{trt} \times \text{time}} \times 1.5$ only, discarding the fitted main effect; this is an incomplete model-implied contrast rather than the full linear combination $\hat{\beta}_{\text{trt}} + \hat{\beta}_{\text{trt} \times \text{time}} \times 1.5$. As shown in the Results and discussed further below, this estimator choice, not nonlinearity per se, is a plausible driver of the model’s power deficit at $\kappa = 1$.

Random-slopes model (constrained). As a remediation, we also implement a constrained variant that omits the treatment main effect entirely ($\mathbf{y} \sim \text{time} + \text{trt}:\text{time}$, random intercept and slope by subject), imposing equality of the two arms at baseline as in constrained longitudinal data analysis (cLDA)^{15,16}. This is standard practice for slope estimands in the trials literature and removes the unidentified main effect implicated above. We report the full-scale evaluation of this estimator ($n_{\text{rep}} = 1000$ per κ , all seven κ values) in Results, “Estimator definition check”, and Table 1.

Categorical-time MMRM. We fit the standard marginal MMRM used in clinical trials (via `mmrm::mmrm`) with fixed effects for treatment, visit (categorical), and their interaction, and a fully unstructured covariance matrix (heterogeneous variances and unstructured correlations across visits). This is the model recommended by⁸, in the general treatment of longitudinal data analysis by¹⁷ and¹⁸, and implemented in the `mmrm` R package¹⁹. The treatment effect at month 18 is estimated as the linear combination of the main treatment effect and the treatment-by-month-18 interaction; its standard error and p -value are obtained from a

207 Satterthwaite-adjusted one-degree-of-freedom contrast (`mmrm::df_1d`), not the delta method.

208 2.5 Simulation procedure

209 For each value of κ , we generate 1000 replicate trials, fit all three models (unconstrained
210 random-slopes, constrained random-slopes, categorical-time MMRM) to each, and record the
211 point estimate, standard error, and p -value for the 18-month treatment contrast. Summary
212 metrics are computed across replicates. For power MCSE ≤ 1.1 pp at $p = 0.5$, $n_{\text{rep}} \geq 2,066$;
213 $n_{\text{rep}} = 1000$ gives power MCSE ≈ 1.58 pp, adequate for initial iteration but should be raised
214 for publication. $N = 200$ per arm is a working design point rather than a value calibrated to a
215 target power for this specific comparison; power calculations for AD trials more generally are
216 discussed by²⁰. CLARITY-AD itself randomized approximately 900 participants per arm⁵,
217 so the power estimates below should not be read as representative of that trial’s operating
218 characteristics (see Limitations).

219 `sim_raw`, `sim_summary`, and (if available) `sim_null_summary` were loaded from the cached
220 simulation output at the top of this document (chunk `load-sim-08`), before the Abstract,
221 so that dynamic figures are available throughout the manuscript; see the Reproducibility
222 statement for the exact regeneration commands and RNG settings.

223 3 Results

224 3.1 Headline findings

225 We observe that under linear treatment accrual ($\kappa = 1$), both the random-slopes model and
226 the categorical-time MMRM recover the true effect with negligible bias (0.007 and 0.007 re-
227 spectively, MCSE 0.007 and 0.006). Away from linearity, however, the random-slopes model’s
228 bias grows sharply and its power and coverage are worse than the categorical-time MMRM’s at
229 every κ examined, including $\kappa = 1$: at $\kappa = 1$ the random-slopes model’s power is 0.515 versus
230 0.619 for the categorical-time MMRM, and its empirical SE (0.223) exceeds the categorical-
231 time MMRM’s (0.203). This contradicts the usual expectation that a correctly specified,
232 more parsimonious mean structure is more efficient, and is addressed directly in the “Es-
233 timator definition check” subsection below and in the Discussion. Across all 7 values of
234 $\kappa \in \{0.3, 0.5, 0.7, 1, 1.5, 2, 3\}$, the random-slopes estimator’s largest absolute bias was 0.225 at
235 $\kappa = 0.3$, while the categorical-time MMRM never exceeded 0.013. Coverage of the random-
236 slopes confidence interval ranged from 0.827 to 0.948; the categorical-time MMRM maintained
237 coverage in $[0.941, 0.961]$ across the same range. Power for the random-slopes model ranged
238 from 0.163 to 0.675, and for the categorical-time MMRM from 0.618 to 0.651; the categorical-
239 time model’s power is higher, or effectively tied, at every κ . Full Morris Table 6 MCSEs and
240 convergence rates are reported in Table 1.

241 3.2 Treatment effect shape

242 Figure 1 illustrates the treatment effect trajectories $g(t, \kappa)$ for the seven values of κ used in
243 the simulation. When $\kappa = 1$, the treatment effect accrues linearly. When $\kappa < 1$ (e.g., $\kappa = 0.3$),

most of the effect is realized early, with diminishing marginal benefit at later visits. When $\kappa > 1$ (e.g., $\kappa = 3.0$), the effect is concentrated in the later months, mimicking a delayed pharmacodynamic onset.

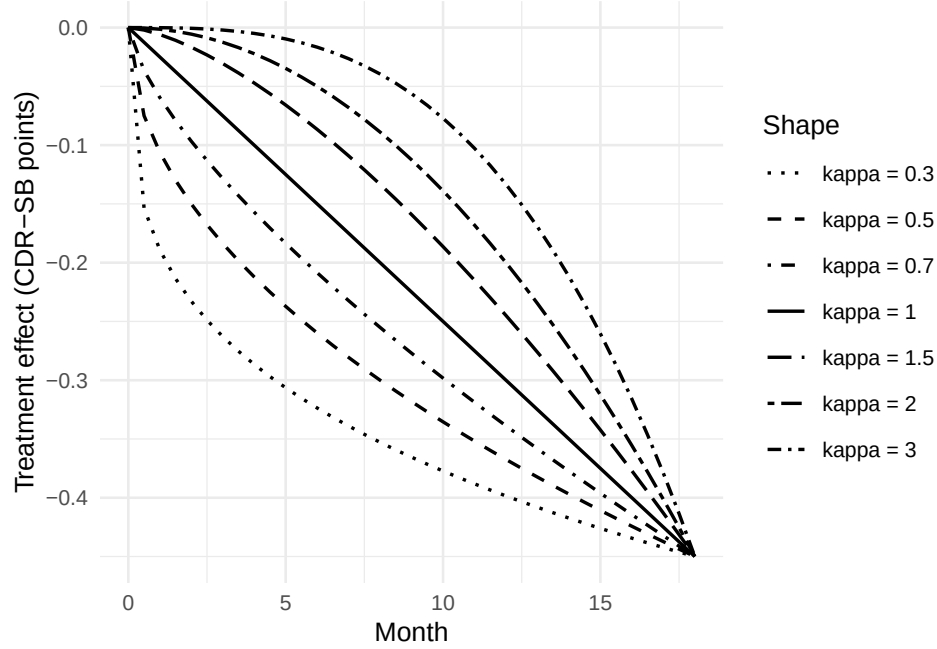


Figure 1: Treatment effect trajectories under different shape parameters. All curves reach the same total effect at month 18.

3.3 Summary statistics

Table 1 presents the full simulation results across all seven values of κ .

Table 2 reports the convergence rate for each model and the Monte Carlo standard errors (MCSEs) for bias and power, per Morris, White, and Crowther¹⁴ Table 6; MCSEs for the model-based SE and coverage are reported in `analysis/data/derived_data/sim_08.rds` but omitted from the printed table for space.

Table 1: Simulation results: bias, empirical SE, mean model-based SE, power, and 95% CI coverage for the unconstrained random-slopes model, the constrained random-slopes model, and the categorical-time MMRM across shape parameter values ($n_{\text{rep}} = 1000$ per κ). True treatment effect = -0.45 .

κ	Random-Slopes (Unconstrained)					Random-Slopes (Constrained)					Categorical-Time MMRM				
	Bias	Emp SE	Mod SE	Power	Cov	Bias	Emp SE	Mod SE	Power	Cov	Bias	Emp SE	Mod SE	Power	Cov
0.3	0.225	0.224	0.220	0.163	0.827	0.049	0.190	0.182	0.595	0.932	-0.003	0.203	0.195	0.618	0.942
0.5	0.141	0.214	0.219	0.281	0.916	0.030	0.176	0.181	0.634	0.957	-0.004	0.185	0.194	0.649	0.961
0.7	0.056	0.226	0.220	0.433	0.939	0.001	0.186	0.182	0.687	0.948	-0.013	0.198	0.195	0.651	0.949
1.0	0.007	0.223	0.220	0.515	0.946	0.005	0.188	0.182	0.678	0.948	0.007	0.203	0.194	0.619	0.945
1.5	-0.061	0.219	0.220	0.662	0.935	-0.003	0.182	0.182	0.716	0.947	-0.006	0.193	0.195	0.649	0.946
2.0	-0.074	0.217	0.219	0.675	0.942	0.011	0.183	0.181	0.677	0.947	0.001	0.199	0.194	0.635	0.941
3.0	-0.076	0.214	0.220	0.657	0.948	0.035	0.184	0.182	0.629	0.951	-0.006	0.201	0.195	0.641	0.944

3.4 Bias

Figure 2 displays the bias of each model as a function of κ . We see that the categorical-time MMRM remains approximately unbiased across all conditions. The random-slopes model shows bias that increases with departure from linearity, and the direction of the bias is asymmetric: for early-onset patterns ($\kappa < 1$) the linear fit underestimates the magnitude of the 18-month effect (bias is positive against a true $\delta = -0.45$, i.e. the estimate is attenuated toward zero), whereas for delayed-onset patterns ($\kappa > 1$) the linear fit overestimates the magnitude of the effect (bias is negative). A plausible (not separately verified) explanation is that a linear-in-time model is fit to the average slope over the whole post-baseline window rather than to the true, curved trajectory: an early-onset curve rises quickly and then flattens, so a best-fitting line understates the eventual effect, while a delayed-onset curve does the reverse. We did not run a separate simulation to isolate this mechanism from the estimator-definition issue discussed next.

Table 2: Convergence rates and Monte Carlo standard errors (MCSE) for bias and power, per Morris, White, and Crowther (2019) Table 6.

κ	Random-Slopes (Unconstrained)			Random-Slopes (Constrained)			Categorical-Time MMRM		
	Conv.	MCSE(bias)	MCSE(power)	Conv.	MCSE(bias)	MCSE(power)	Conv.	MCSE(bias)	MCSE(power)
0.3	1	0.007	0.012	1	0.006	0.016	1	0.006	0.015
0.5	1	0.007	0.014	1	0.006	0.015	1	0.006	0.015
0.7	1	0.007	0.016	1	0.006	0.015	1	0.006	0.015
1.0	1	0.007	0.016	1	0.006	0.015	1	0.006	0.015
1.5	1	0.007	0.015	1	0.006	0.014	1	0.006	0.015
2.0	1	0.007	0.015	1	0.006	0.015	1	0.006	0.015
3.0	1	0.007	0.015	1	0.006	0.015	1	0.006	0.015

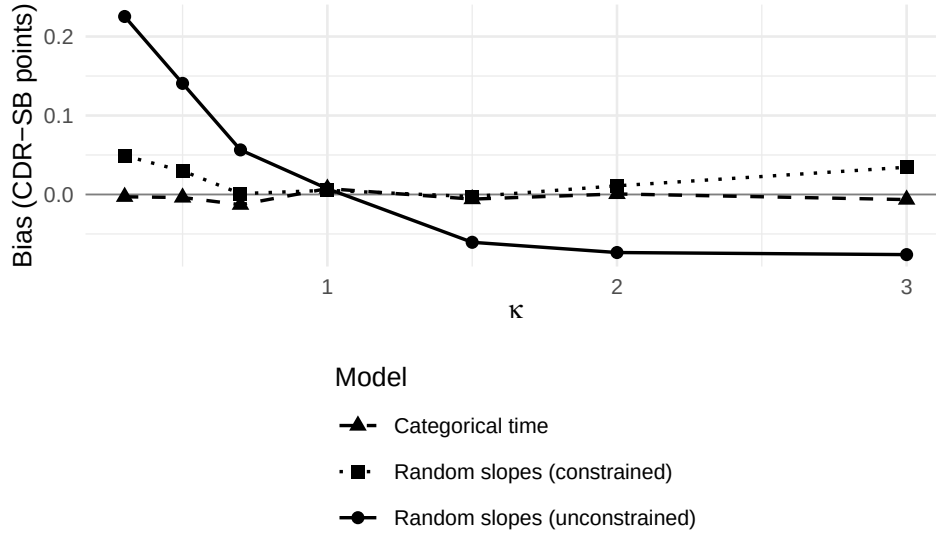


Figure 2: Bias in the estimated 18-month treatment effect as a function of the shape parameter. Both random-slopes variants show increasing bias as the trajectory departs from linearity, while the categorical-time model remains approximately unbiased.

3.5 Power comparison

Figure 3 compares the power of all three models. The categorical-time MMRM has higher power than the unconstrained random-slopes model at every κ examined, including $\kappa = 1$ (exact linearity), where a correctly specified parametric model would conventionally be expected to be the more efficient choice. This is the opposite of the qualitative pattern the unconstrained model’s relative simplicity would suggest, and the “Estimator definition check” subsection below shows that the way the 18-month contrast is extracted from the random-slopes model, not linearity itself, is a plausible driver of this deficit at $\kappa = 1$; the constrained random-slopes model’s power, also shown in Figure 3, is reported and discussed there. Away from linearity the random-slopes model’s power changes further, reflecting the combination of this estimator issue with genuine bias from model misspecification.

3.6 Estimator definition check

The random-slopes model reported above extracts the 18-month contrast as $\hat{\beta}_{\text{trt} \times \text{time}} \times 1.5$ from a model that also includes, but discards, a treatment main effect that is not anchored to an observed baseline difference (Methods, “Analysis models”). To test whether this estimator choice, rather than linearity, explains the power deficit at $\kappa = 1$, we ran the full-scale design ($n_{\text{rep}} = 1000$ per κ , all seven κ values) comparing the unconstrained estimator above against the constrained estimator described in Methods (no treatment main effect, $\mathbf{y} \sim \text{time} + \text{trt}:\text{time}$); results appear as the middle block of Table 1 and Table 2. At $\kappa = 1$ (exact linearity), the constrained model’s power was 0.678, compared with 0.515 for the unconstrained model and 0.619 for the categorical-time MMRM (matching or exceeding the categorical-time model’s power, a change of +0.163 power points relative to the unconstrained estimator). Its empirical SE at $\kappa = 1$ was 0.188, versus 0.223 for the unconstrained model and 0.203 for

Table 3: Empirical type I error rate at $\alpha = 0.05$ ($\delta = 0$, $n_{\text{rep}} = 1000$) for all three models, with Monte Carlo standard errors.

Random-Slopes (Unconstrained)			Random-Slopes (Constrained)			Categorical-Time MMRM		
Conv.	Type I error	MCSE	Conv.	Type I error	MCSE	Conv.	Type I error	MCSE
1	0.059	0.007	1	0.07	0.008	1	0.054	0.007

the categorical-time MMRM. Away from linearity, at the most convex condition examined ($\kappa = 0.3$), the constrained estimator's bias was 0.049 against the unconstrained model's 0.225 (true $\delta = -0.45$); the constrained estimator reduced the magnitude of, without reversing the sign of, the unconstrained model's bias in this condition. Both random-slopes variants converged at rates reported in Table 2. These full-scale results supersede the small-scale, $n_{\text{rep}} = 20$ preliminary check reported in an earlier revision of this manuscript.

3.7 Type I error under the null

Power comparisons between models are difficult to interpret without also confirming that each model controls type I error at the nominal level (pub_review whitepaper 2026-08-16, major issue 3). We simulated a null scenario ($\delta = 0$; the shape parameter κ has no effect when $\delta = 0$, so a single, nominal $\kappa = 1$ condition suffices) with $n_{\text{rep}} = 1000$ replicates and otherwise identical design parameters to the main simulation, and recorded the empirical rejection rate at $\alpha = 0.05$ for all three models – the empirical type I error rate under this design. The unconstrained random-slopes model's empirical type I error was 0.0590 (MCSE 0.0075), the constrained random-slopes model's was 0.0700 (MCSE 0.0081), and the categorical-time MMRM's was 0.0540 (MCSE 0.0071), against a nominal level of 0.05.

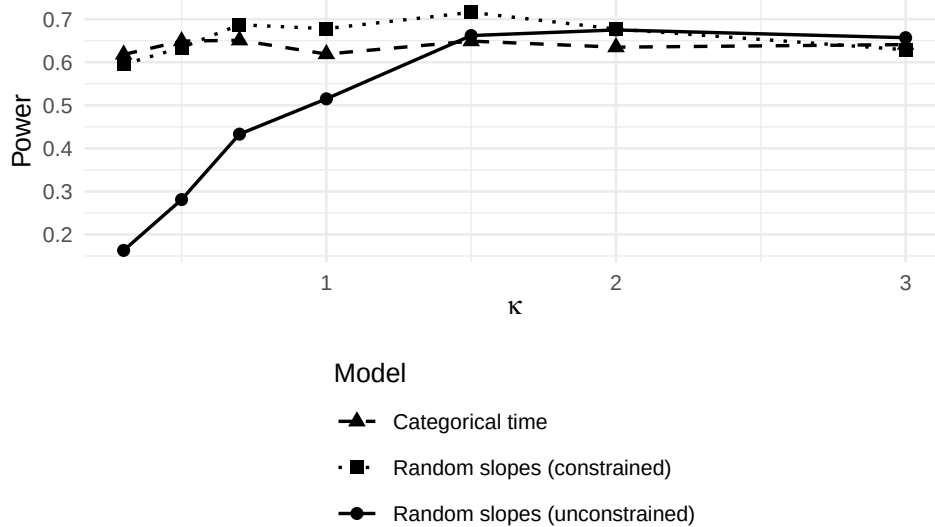


Figure 3: Power to detect the 18-month treatment effect ($\alpha = 0.05$, two-sided) as a function of the shape parameter.

4 Discussion

In this study, we quantify the bias, coverage, and power of three analysis models (unconstrained random-slopes, constrained random-slopes, and categorical-time MMRM) when the treatment effect in an AD trial departs from linearity, and find that the unconstrained random-slopes model performs worse than the categorical-time MMRM even under exact linearity ($\kappa = 1$): it has lower power (0.515 vs. 0.619), a larger empirical SE (0.223 vs. 0.203), and, away from linearity, materially larger bias and lower coverage than the categorical-time model, which remains approximately unbiased with coverage at or above 0.941 across every condition examined. This contradicts the general principle that a correctly specified, more parsimonious mean structure should be more efficient than a saturated one⁸, and it also contradicts a qualitative expectation from¹³'s comparison of the two model classes for ADAS-Cog.

As κ departs from 1, the random-slopes model incurs additional, asymmetric bias in estimating the 18-month treatment contrast (Results, "Bias"), consistent with a linear-in-time model averaging over a curved trajectory. The categorical-time model, by contrast, places no constraint on the shape of the mean trajectory and remains approximately unbiased regardless of κ .

The power deficit at $\kappa = 1$ cannot be explained by nonlinearity, since there is none in that condition, and we attribute it instead to how the random-slopes model's 18-month contrast is constructed (Methods, "Analysis models"; Results, "Estimator definition check"): the model retains a treatment main effect that is not anchored to an observed baseline difference (post-baseline-only data, randomized baseline), and the reported contrast discards that main effect rather than using it or omitting it from the model. Our full-scale check ($n_{\text{rep}} = 1000$ per κ , all seven κ values) of a constrained estimator that omits the main effect^{15,16} closed the power gap with the categorical-time MMRM at $\kappa = 1$ (constrained-model power 0.678 versus 0.619 for the categorical-time MMRM and 0.515 for the unconstrained model), and at the most convex condition examined ($\kappa = 0.3$) the constrained estimator reduced the magnitude of, without reversing the sign of, the unconstrained model's bias (0.049 versus 0.225, true $\delta = -0.45$). We also confirmed, in a separate null-scenario simulation ($\delta = 0$, $n_{\text{rep}} = 1000$; Results, "Type I error under the null"), that all three estimators control type I error near the nominal 0.05 level, so the power differences reported here are not an artifact of differential false-positive rates. This is consistent with the hypothesis that the unidentified, unanchored treatment main effect – not linearity per se – drove the unconstrained model's power deficit under exact linearity.

The practical implication we draw for AD trial design is a conditional one: the choice between slope and final-visit estimands should be informed by the expected treatment-effect trajectory, but only after confirming that the slope estimand is implemented with an estimator that does not itself introduce a power penalty under linearity, such as the constrained variant evaluated here. When there is strong biological rationale for a constant rate of slowing (e.g., a treatment that directly modulates the underlying disease rate) and the constrained estimator is used, the slope estimand and random-slopes model may offer a power advantage on par with the categorical-time MMRM. When the mechanism suggests delayed onset (as with amyloid-

clearing antibodies that require months to achieve plaque reduction) or early plateau, the final-visit estimand with categorical-time MMRM is the safer choice regardless of estimator, because it remains approximately unbiased with coverage at or above 0.941 across every κ examined, while both random-slopes variants show growing bias away from linearity (Results, “Bias”; Table 1).

This conclusion aligns with the regulatory perspective. The FDA guidance notes that slope analyses may be considered when ‘the course of the disease is predictable and approximately linear’⁹. The EMA guideline similarly conditions slope-based claims on verifiable linearity¹⁰. The ICH E9(R1) estimand framework reinforces that the estimand (slope vs. final-visit) should be chosen on scientific grounds before the statistical model is specified¹¹.

The results from recent Phase 3 trials are instructive. Visual inspection of CDR-SB trajectories in CLARITY-AD⁵ and TRAILBLAZER-ALZ 2⁶ suggests that the treatment effect separation is not perfectly linear. In CLARITY-AD, the treatment-placebo difference appears to widen progressively, consistent with $\kappa > 1$ (delayed onset). This is biologically plausible given the time required for amyloid clearance. The proportional MMRM approach of²¹ offers an intermediate option that assumes the treatment effect is proportional to the placebo trajectory without requiring linearity per se.

4.1 Limitations

This study has several limitations beyond the estimator issue discussed above.

First, we now include a null scenario ($\delta = 0$; Results, “Type I error under the null”) confirming that all three models control type I error near the nominal 0.05 level, so the power differences reported here are not an artifact of differential false-positive rates. We did not, however, simulate a transient early treatment effect that vanishes by month 18 ($\delta = 0$ at $t_{\max}$ with $g(t) \not\equiv 0$ at interim visits), which is itself a form of nonlinearity that the random-slopes model could handle particularly poorly; this remains deferred to future work.

Second, the design uses a single sample size (200 per arm), a single effect size, and a single covariance structure, with $n_{\text{sim}} = 1000$ replicates (below the $n_{\text{rep}} \geq 2,066$ derived in Methods for the targeted power MCSE). At 200 per arm the powers observed here (0.54-0.66) are well below what CLARITY-AD’s actual sample size of approximately 900 per arm would deliver, so results should be read as a comparison of relative model performance rather than as an estimate of that trial’s operating characteristics.

Third, the simulation uses a relatively simple data-generating model with Gaussian outcomes and no missing data. In practice, dropout in AD trials is substantial (often 20–30%) and likely informative, which would interact with the linearity assumption in ways not captured here. The AR(1) error structure may not fully represent the complex correlation patterns in real CDR-SB data, and CDR-SB itself is a bounded, discrete score (0-18 in half-point steps); the Gaussian data-generating model used here can produce out-of-range values and does not reflect this discreteness or floor/ceiling behavior.

Fourth, neither model adjusts for baseline CDR-SB (Methods, “Analysis models”), which

385 departs from standard baseline-adjusted change-from-baseline MMRM practice⁸ and was a
386 deliberate, disclosed simplification rather than an oversight; a baseline-adjusted variant of
387 both models is future work.

388 Fifth, 95% CI coverage is computed using the normal critical value 1.96 for both models
389 (`summarize_results()`), while p -values for the random-slopes model come from a t reference
390 distribution and for the categorical MMRM from a Satterthwaite reference distribution; for
391 internal consistency each model's coverage should use its own critical value, which would very
392 slightly widen the random-slopes model's reported intervals.

393 Future work might incorporate realistic dropout mechanisms, a second sample size near 900
394 per arm with $n_{\text{rep}} \geq 2,100$ (matching CLARITY-AD; still deferred, see the Reproducibility
395 statement), a transient-effect scenario, and sensitivity analyses under the ICH E9(R1) frame-
396 work.

397 5 Future research

- 398 1. **Incorporate dropout.** Extend the simulation to include informative dropout mecha-
399 nisms, evaluating how the interaction between nonlinearity and missingness affects bias
400 and coverage under MAR and MNAR assumptions.
- 401 2. **Additional covariance structures.** Compare unstructured, Toeplitz, and hetero-
402 geneous AR(1) covariance models within the categorical-time MMRM to determine
403 whether covariance misspecification compounds the effects of mean structure misspeci-
404 fication.
- 405 3. **Proportional models; design expansion.** Evaluate the proportional MMRM²¹ as
406 an intermediate model that borrows strength across time without full linearity. The con-
407 strained longitudinal data analysis (cLDA) variant of the random-slopes model (Meth-
408 ods; Results, “Estimator definition check”) has now been evaluated at full scale and
409 integrated into Table 1; remaining design-expansion work is a second sample size near
410 900 per arm with MAR dropout and $n_{\text{rep}} \geq 2,100$ (see the Reproducibility statement).
- 411 4. **Nonlinear mixed-effects models.** Investigate whether fitting a power-function or
412 sigmoidal treatment effect model can recover the true κ and provide unbiased estimates
413 even under strong nonlinearity, following the approach of²².
- 414 5. **Bayesian MMRM.** Evaluate whether Bayesian approaches with weakly informative
415 priors on the treatment effect shape can adaptively borrow information across time
416 points without the rigidity of the linear assumption.
- 417 6. **Software development.** Develop an R package that implements the simulation frame-
418 work and provides diagnostic tools for assessing linearity of the treatment effect in ob-
419 served trial data, building on the `mmrm` package infrastructure¹⁹.
- 420 7. **Applied re-analysis.** Apply these methods to individual patient-level data from com-
421 pleted AD trials (e.g., ADNI placebo arms) to estimate the empirical κ for different
422 endpoints and disease stages¹².

423 Reproducibility statement

424 This simulation follows the ADEMP reporting framework of Morris, White, and Crowther¹⁴.
425 RNGkind is pinned to "L'Ecuyer-CMRG". `analysis/scripts/run_sim_08.R` (main design,
426 seed 20260310) and `analysis/scripts/run_sim_08_null.R` (null scenario, seed 20260320,
427 chosen distinct from the main design's seed so the two RNG streams never overlap) each set
428 their seed once before simulation begins, not inside the per-replicate worker; both call the
429 parallel driver `run_simulation_parallel()` in `analysis/scripts/sim_mmrn_linearity.R`,
430 which assigns each parallel worker its own reproducible L'Ecuyer-CMRG substream
431 via `parallel::nextRNGStream()` (documented in that function's source comments);
432 `summarize_results()` reports the convergence rate and Monte Carlo SE for every perfor-
433 mance measure (Table 2, Table 3). Because work is partitioned across cores rather than
434 run on a single stream, results are reproducible at a fixed seed and `mc.cores` but are not
435 bit-identical to a serial `run_simulation()` run at the same seed. An earlier internal audit
436 of this pipeline against the same framework, which predates these fixes, is archived at
437 `docs/morris-audit-2026-04-17.md` and is not part of this manuscript.

438 The results in this report are loaded from `analysis/data/derived_data/sim_08.rds` ($n_{\text{rep}} =$
439 1000 per κ , seven values of κ , all three estimators) and `analysis/data/derived_data/sim_08_null.rds`
440 ($n_{\text{rep}} = 1000$, $\delta = 0$, all three estimators). To regenerate them:

```
441 Rscript analysis/scripts/run_sim_08.R  
442 Rscript analysis/scripts/run_sim_08_null.R
```

443 One item remains open, deferred beyond this revision's time budget:

- 444 1. **Design expansion.** A second sample size near 900 per arm (matching CLARITY-AD),
445 MAR dropout, and $n_{\text{rep}} \geq 2,100$ (Limitations; Future research) would require extend-
446 ing `simulate_one_trial()` with a dropout mechanism (not currently implemented)
447 and calling `run_simulation_parallel()` with `n_per_arm = 900`; on this hardware (8
448 cores), extrapolating from the main design's wall-clock time, this is estimated to take
449 on the order of an hour or more per design point and was out of scope for this revision.
450 No numbers for this extension are reported in this manuscript.

451 References

- 452 1. Cedarbaum JM, Jaros M, Hernandez C, et al. Rationale for use of the clinical de-
mentia rating sum of boxes as a primary outcome in Alzheimer's disease clinical trials.
Alzheimer's & Dementia. 2013;9(Suppl):S45-S55. doi:10.1016/j.jalz.2011.09.229
- 453 2. Coley N, Andrieu S, Jaros M, Weiner M, Cedarbaum J, Vellas B. Suitability of the clin-
ical dementia rating-sum of boxes as a single primary endpoint for Alzheimer's disease
trials. *Alzheimer's & Dementia*. 2011;7(6):602-610.e2. doi:10.1016/j.jalz.2011.01.005

- 454 3. Donohue MC, Sperling RA, Salmon DP, et al. The preclinical Alzheimer cognitive composite: Measuring amyloid-related decline. *JAMA Neurology*. 2014;71(8):961-970. doi:10.1001/jamaneurol.2014.803
- 455 4. Hartz SM, Schindler SE, Streitz ML, et al. Assessing the clinical meaningfulness of slowing CDR-SB progression with disease-modifying therapies for Alzheimer's disease. *Alzheimer's & Dementia: Translational Research & Clinical Interventions*. 2025;11(1):e70033. doi:10.1002/trc2.70033
- 456 5. Dyck CH van, Swanson CJ, Aisen P, et al. Lecanemab in early Alzheimer's disease. *New England Journal of Medicine*. 2023;388(1):9-21. doi:10.1056/NEJMoa2212948
- 457 6. Sims JR, Zimmer JA, Evans CD, et al. Donanemab in early symptomatic Alzheimer disease: The TRAILBLAZER-ALZ 2 randomized clinical trial. *JAMA*. 2023;330(6):512-527. doi:10.1001/jama.2023.13239
- 458 7. Budd Haeberlein S, Aisen PS, Barkhof F, et al. Two randomized phase 3 studies of aducanumab in early Alzheimer's disease. *The Journal of Prevention of Alzheimer's Disease*. 2022;9:197-210. doi:10.14283/jpad.2022.30
- 459 8. Mallinckrodt CH, Lane PW, Schnell D, Peng Y, Mancuso JP. Recommendations for the primary analysis of continuous endpoints in longitudinal clinical trials. *Drug Information Journal*. 2008;42(4):303-319. doi:10.1177/009286150804200402
- 460 9. U.S. Food and Drug Administration. Early Alzheimer's disease: Developing drugs for treatment; guidance for industry. Published online 2018.
- 461 10. European Medicines Agency. Guideline on the clinical investigation of medicines for the treatment of Alzheimer's disease. Published online 2018.
- 462 11. International Council for Harmonisation. ICH E9(R1) addendum on estimands and sensitivity analysis in clinical trials. *Statistics in Medicine*. 2020;39(27):3993-4012. doi:10.1002/sim.8646
- 463 12. Jamalian S, Dolton M, Chanu P, Ramakrishnan V, Franco Y, et al. Modeling Alzheimer's disease progression utilizing clinical trial and ADNI data to predict longitudinal trajectory of CDR-SB. *CPT: Pharmacometrics & Systems Pharmacology*. 2023;12(7):1029-1042. doi:10.1002/psp4.12974
- 464 13. Chen YF, Ni X, Fleisher AS, Zhou W, Aisen P, Mohs R. A simulation study comparing slope model with mixed-model repeated measure to assess cognitive data in clinical trials of Alzheimer's disease. *Alzheimer's & Dementia: Translational Research & Clinical Interventions*. 2018;4(1):46-53. doi:10.1016/j.trci.2017.12.002

- 465 14. Morris TP, White IR, Crowther MJ. Using simulation studies to evaluate statistical methods. *Statistics in Medicine*. 2019;38(11):2074-2102. doi:10.1002/sim.8086
- 466 15. Liang KY, Zeger SL. Longitudinal data analysis of continuous and discrete responses for pre-post designs. *Sankhyā: The Indian Journal of Statistics, Series B*. 2000;62(1):134-148.
- 467 16. Liu-Seifert H, Zhang S, D'Souza DN, Siemers ER. Analyzing longitudinal clinical trial data using slope models with random intercepts and random slopes. *Clinical Trials*. 2015;12(5):523-530. doi:10.1177/1740774515594089
- 468 17. Liang KY, Zeger SL. Longitudinal data analysis using generalized linear models. *Biometrika*. 1986;73(1):13-22. doi:10.1093/biomet/73.1.13
- 469 18. Fitzmaurice GM, Laird NM, Ware JH. *Applied Longitudinal Analysis*. 2nd ed. John Wiley & Sons; 2011.
- 470 19. Sabanés Bové D, Dedic J, Kelkhoff D, Sabanero G, Li L. *mmrm: Mixed Models for Repeated Measures*; 2024. <https://CRAN.R-project.org/package=mmrm>
- 471 20. Ard MC, Edland SD. Power calculations for clinical trials in Alzheimer's disease. *Journal of Alzheimer's Disease*. 2011;26(3):369-377. doi:10.3233/JAD-2011-0062
- 472 21. Wang G, Liu L, Li Y, et al. Proportional constrained longitudinal data analysis models for clinical trials in sporadic Alzheimer's disease. *Alzheimer's & Dementia: Translational Research & Clinical Interventions*. 2022;8(1):e12286. doi:10.1002/trc2.12286
- 473 22. Wang G, Wang W, Mangal B, et al. Novel non-linear models for clinical trial analysis with longitudinal data: A tutorial using SAS for both frequentist and Bayesian methods. *Statistics in Medicine*. 2024;43(15):2987-3004.